Incorporation of Evidence into Immune Epitope Assays

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ABSTRACT

The Immune Epitope Database (IEDB) is a freely available, National Institute of Allergy and Infectious Diseases (NIAID) funded resource presenting published experimental data on the adaptive immune response to more than 1.6 million antibody and T cell epitopes studied in the context of infectious disease, allergy, autoimmunity, and transplantation. Epitopes are the portions of an antigen that are recognized by antibodies and T cell receptors. Epitopes are key to understanding healthy and abnormal immune responses and are crucial in the development of vaccines and therapeutics. T cells recognize epitopes when presented by specific major histocompatibility complex (MHC) molecules. Which specific MHC molecule can present an epitope is critically important to immunologists, however, the published data is not always precise. Different experimental methods result in varying degrees of knowledge. In order to clarify this information to our users, the IEDB collaborated with the Evidence and Conclusion Ontology (ECO) to standardize the methodology and precision of MHC presentation and restriction information. ECO contains terms that describe types of evidence and assertion methods used to capture the evidence supporting biological assertions. Formalizing evidence allows tracking of annotation provenance, the establishment of quality control measures within datasets, and gives us the ability to query datasets by evidence.
We itemized the various methods used by the publications curated in the IEDB and analyzed the relationships between the methodologies used and the types of assertion the researchers were making to develop an ECO term template. In the process of creating the new ECO terms, we utilized and contributed to other ontologies as needed. The IEDB incorporated these terms into our website design and display. Additionally, the logical definitions were utilized to validate the data in the IEDB. We present our collaborative use case as an example of the mutual benefits of ontologies and databases working together.

INTRODUCTION

The IEDB contains 3 main types of experimental assays used to measure epitope specific adaptive immune responses; B cell, T cell, and MHC ligand assays. The goal of this project was to formalize evidence of MHC restriction and/or presentation, which applies to T cell assays and to a subset of the MHC ligand assays, specifically the MHC ligand elution assays. MHC molecules are important to our understanding of the adaptive immune response, as they are differentially expressed in different populations and different MHC are capable of presenting different epitopes. These differences can mean that one person is able to mount an adequate immune response to a pathogen with very few symptoms, while another becomes severely ill. As shown in Figure 1A, Antigen Presenting Cells (APC) process antigens into smaller pieces, some of which can be presented by/bound to specific MHC molecules. The specific MHC molecule that an epitope is eluted out of is considered as being capable of presenting that epitope. T cells recognize epitopes when presented by specific major histocompatibility complex (MHC) molecules. Which specific MHC molecule can present an epitope is critically important to immunologists, however, the published data is not always precise. Different experimental methods result in varying degrees of knowledge. In order to clarify this information to our users, the IEDB collaborated with the Evidence and Conclusion Ontology (ECO) to standardize the methodology and precision of MHC presentation and restriction information. ECO contains terms that describe types of evidence and assertion methods used to capture the evidence supporting biological assertions. Formalizing evidence allows tracking of annotation provenance, the establishment of quality control measures within datasets, and gives us the ability to query datasets by evidence.
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RESOURCES

Immune Epitope Database (IEDB)

- Experimental assays derived from many different publications using up to 400 fields per assay
- > 1.6 million antibody or T cell assays
- Infectious disease, allergy, autoimmunity, and transplantation contexts
- iedb.org

Evidence and Conclusion Ontology (ECO)

- Ontological terms to describe types of evidence and assertion methods
- Allows tracking of annotation provenance, establishment of quality control measures, and evidence-based data mining
- evidenceontology.org

REFERENCES

IEDB: Vita R, Mahajan S, Overton JA, Dhanda SK, Martini S, Cantrell JR, Wheeler DK, Sette A, Peters B. The Immune Epitope Database (IEDB): 2018 update. Nucleic Acids Res. 2019 Jan 8;47(D1):D339-D343. doi: 10.1093/nar/gky1006. PMID: 30357391; PMCID: PMC6324067.

ECO: Nadendla S, Jackson R, Munro J, Quaglia F, Mészáros B, Olley D, Hobbs ET, Goralski SM, Chibucos M, Mungall CJ, Tosatto SCE, Erill I, Giglio MG. ECO: the Evidence and Conclusion Ontology, an update for 2022. Nucleic Acids Res. 2022 Jan 7;50(D1):D1515-D1521. doi: 10.1093/nar/gkab1025. PMID: 34986598; PMCID: PMC8728134.

OBI: Bandrowski A, Brinkman R, Brochhausen M, Brush MH, Bug B, Chibucos MC, Clancy K, Courtot M, Derom D, Dumontier M, Fan L, Fostel J, Fragoso G, Gibson F, Gonzalez-Beltran A, Haendel MA, He Y, Heiskanen M, Hernandez-Boussard T, Jensen M, Lin Y, Lister AL, Lord P, Malone J, Manduchi E, McGee M, Morrison N, Overton JA, Parkinson H, Peters B, Rocca-Serra P, Ruttenberg A, Sansone SA, Scheuermann RH, Schober D, Smith B, Soldatova LN, Stoeckert CJ Jr, Taylor CF, Tornial C, Turner JA, Vita R, Whetzel PL, Zheng J. The Ontology for Biomedical Investigations. PLoS One. 2016 Apr 29;11(4):e0154556. doi: 10.1371/journal.pone.0154556. PMID: 27128319; PMCID: PMC4851331.

MRO: Vita R, Overton JA, Seymour E, Sidney J, Kaufman J, Tallmadge RL, Ellis S, Hammond J, Butcher GW, Sette A, Peters B. An ontology for major histocompatibility restriction. J Biomed Semantics. 2016 Jan 11;7:1. doi: 10.1186/s13326-016-0045-5. PMID: 26759709; PMCID: PMC4709943.

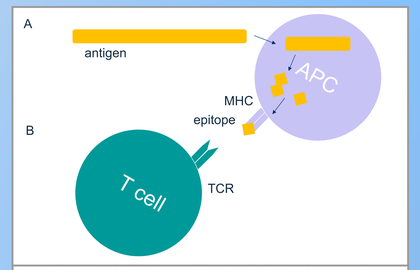


Figure 1A. MHC Ligand Elution Assays. Antigen Presenting Cells (APC) process antigens into smaller pieces, some of which can be presented by/bound to specific MHC molecules. The specific MHC molecule that an epitope is eluted out of is considered as being capable of presenting that epitope.
Figure 1B. MHC Restriction in T cell Assays. Antigen Presenting Cells (APC) process antigens and present epitopes by specific MHC molecules to T cells which recognize them using using their T cell receptors (TCR). The specific MHC molecule capable of presenting an epitope to a T cell is known as the epitope's MHC Restriction, as measured by a T cell response.

METHODS

We reviewed all experimental procedures used in the 4,063,930 MHC ligand elution assays and 539,898 T cell assays curated in the IEDB where MHC presentation and/or restriction was relevant. To describe the various methods to assert MHC restriction or presentation, we assembled the 32 methodologies used to identify the identity of the MHC molecule(s) that an epitope was presented by, shown in Table 1. Similarly, we compiled a list of a list of 10 T cell assay MHC restriction determination methods in use by the IEDB, as shown in Table 2.
We then analyzed the relationship between the methodology used and the type of assertion the researchers were making in the publications being curated to develop a design pattern for the new ECO terms. We utilized OBI and MRO terms to model the methodologies in the evidence term lists and create a term template as shown in Table 3. This template was used to model the IEDB terms and resulted in a total of 144 new ECO terms, including organizational nodes and terms required to create new concepts in ECO, as shown in Figure 2.
Once the new ECO terms were publicly released, the IEDB incorporated them into its database in the form of the query interface, web links, and data exports, as shown in Figure 3. Validation rules were constructed to capitalize on the ontological axioms. The rules identified errors in the IEDB, which were corrected, as demonstrated in Figure 4. For example, if the antibody used to determine MHC presentation evidence is specific for a particular MHC molecule, the assay should not have a different MHC molecule in the MHC database field.

Table 1

IEDB Evidence Code	Type
Antibody purification	Presentation
Antibody purification - 10.3.6	Presentation
Antibody purification - 14.4.4	Presentation
Antibody purification - 28-14-85	Presentation
Antibody purification - 34-5-85	Presentation
Antibody purification - 4B7/8	Presentation
Antibody purification - AG2.42.7	Presentation
Antibody purification - B123.2	Presentation
Antibody purification - B22-249.R1	Presentation
Antibody purification - B7/21	Presentation
Antibody purification - B8-24-3	Presentation
Antibody purification - B87.2	Presentation
Antibody purification - DK1G8	Presentation
Antibody purification - DT9	Presentation
Antibody purification - HB145	Presentation
Antibody purification - HB180	Presentation
Antibody purification - HB195	Presentation
Antibody purification - IVD12	Presentation
Antibody purification - L243	Presentation
Antibody purification - LB3.1	Presentation
Antibody purification - M1/42	Presentation
Antibody purification - M5/114	Presentation
Antibody purification - ME1	Presentation
Antibody purification - MK06	Presentation
Antibody purification - SF1-1.1.1	Presentation
Antibody purification - SPVL3	Presentation
Antibody purification - Tu39	Presentation
Antibody purification - W6/32	Presentation
Antibody purification - Y-3	Presentation
Antibody purification - YB3F	Presentation
Inferred by motif or alleles present	Presentation
Single allele present	Presentation

Table 2

IEDB Evidence Code	Type
Cited reference	Restriction
MHC binding assay	Restriction
MHC binding prediction	Restriction
Not determined	Restriction
Statistical association	Restriction
T cell assay -Biological process measured	Restriction
T cell assay -MHC subset identification	Restriction
T cell assay -Mismatched MHC molecules	Restriction
T cell assay -Single MHC type present	Restriction
T cell assay -T cell subset identification	Restriction

Table 1. MHC Presentation Evidence. The list of methods used to determine the identity of the MHC molecule presenting a specific epitope found in the curated literature.
Table 2. MHC Restriction Evidence. The list of evidence codes in use by the IEDB to represent the methods used to determine MHC restriction in T cell assays.

rdfs label	textual definition	eco parent	dbxref	C is_specified_output_of some %	and has part
antibody-specificity based MHC presentation evidence	A type of protein binding evidence that employs specific antibodies to determine that a specific MHC protein complex or group of MHC protein-complexes presented a ligand.	protein binding evidence	Antibody purification	assay measuring MHC ligand processing and presentation	antibody purification of MHC protein complex
antibody 10.3.6-specificity based MHC presentation evidence	A type of antibody-specificity based MHC presentation evidence that employs antibody 10.3.6 to determine that the H2-IA protein complex presented a ligand.	antibody-specificity based MHC restriction evidence	Antibody purification - 10.3.6		antibody purification of H2-IA protein complex
protein expression-based MHC presentation evidence	A type of protein expression evidence that employs MHC molecule motif or alleles present to determine that a specific MHC molecule or class of MHC molecules presented a ligand.	protein expression evidence	Inferred by motif or alleles present	assay measuring MHC ligand processing and presentation	major histocompatibility typing assay
specific MHC protein expression-based MHC presentation evidence	A type of protein expression-based MHC presentation evidence that employs knowledge of a single specific MHC molecule's presence to determine that a specific MHC molecule presented a ligand.	protein expression-based MHC presentation evidence	Single allele present	assay measuring MHC ligand processing and presentation	
MHC:ligand binding based MHC restriction evidence	A type of protein binding evidence that employs an assay measuring binding of an MHC molecule to a ligand to determine that T cells can recognize the ligand when presented by that MHC molecule	protein binding evidence	MHC binding assay	assay measuring binding of a MHC:ligand complex	
MHC complex binding prediction based MHC restriction evidence	A type of computational evidence that employs a computational tool to predict MHC complex binding to a ligand to infer that T cells can recognize the ligand when presented by that MHC molecule	computational evidence	MHC binding prediction	mhc binding prediction	
inference of MHC restriction based on shared MHC expression	A type of inferential evidence that employs the association of specific MHC molecule expression by a subject population with a T cell response to infer that T cells can recognize the ligand when presented by that MHC molecule	inferential evidence	Statistically inferred by motif or alleles present	T cell epitope assay	
T cell epitope assay type based MHC restriction evidence	A type of direct assay evidence that employs the type of T cell epitope assay to determine that a specific ligand is restricted by a MHC molecule or class of MHC molecules.	direct assay evidence	T cell assay - Biological process measured	T cell epitope assay	
MHC-specific blocking based MHC restriction evidence	A type of protein binding evidence that employs the specific blocking of MHC molecules to determine that a ligand is restricted by that MHC molecule or class of MHC molecules.	protein binding evidence	T cell assay -MHC subset identification	T cell epitope assay	
specific MHC molecule based MHC restriction evidence	A type of direct assay evidence that employs the use of a specific MHC molecule in a T cell response assay to determine that a specific ligand is restricted by that MHC molecule.	direct assay evidence	T cell assay - Single MHC type present	T cell epitope assay	
T cell type based MHC restriction evidence	A type of direct assay evidence that employs a specific T cell type (CD4+ or CD8+) to determine that a ligand is restricted by a class of MHC molecules.	direct assay evidence	T cell assay -T cell subset identification	T cell epitope assay	

Table 3. Evidence Modeling Design Pattern.
For each term, we worked together to devise a label and textual definition, its place in the ECO hierarchy, and determined what process the evidence is a result of. Some of the evidence methods involved a specific experimental assay type (OBI). This table shows some example terms.

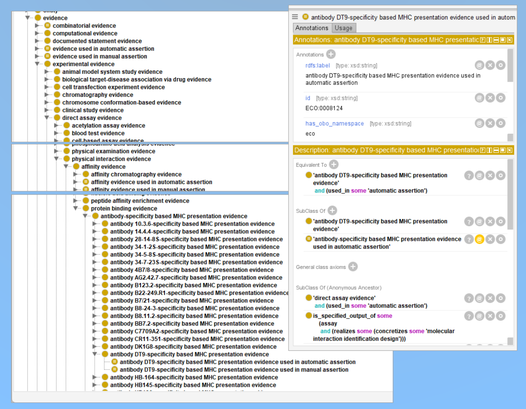
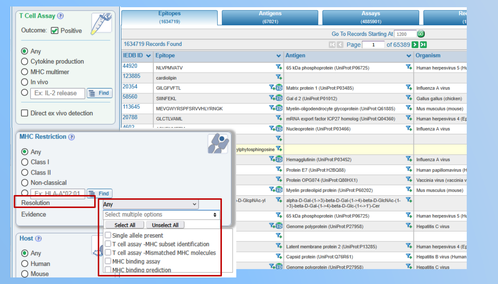


Figure 2. New ECO Terms.
Utilizing the term template, ECO added 144 new terms, which includes organizational nodes and the creation of both manual assertion and automatic assertion terms for each type of evidence.



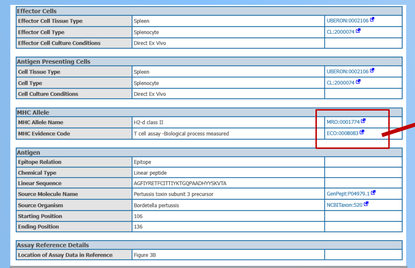
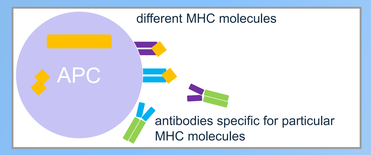


Figure 3. IEDB Incorporates Evidence. The IEDB added a search interface for MHC evidence and links for each ECO evidence code used, to inform users of the details of the evidence methodology used to determine the MHC presentation or restriction in each experimental assay.



antibody DT9-specificity based MHC presentation evidence (ECO:0008057) has_part antibody purification of HLA-C protein complex (OBI:0003623)
antibody purification of HLA-C protein complex (OBI:0003623) is_a antibody purification of human MHC class I protein complex and has specified output some HLA-C protein complex (MRO:0001149)
HLA-C protein complex (MRO:0001149) is_a MHC class I protein complex and has part some HLA-C chain (MRO:0000616)
HLA-C*04:01 protein complex (MRO:0001154) is_a MHC class I protein complex and has_part some HLA-C*04:01 chain (MRO:0000621)
HLA-C*04:01 chain (MRO:0000621) is_a HLA-C chain (MRO:0000616)
HLA-A*02:01 protein complex (MRO:0001007) is_a MHC class I protein complex and has_part some HLA-A*02:01 chain (MRO:0000473)
HLA-A*02:01 chain (MRO:0000473) is_a HLA-A chain (MRO:0000471)
HLA-A chain (MRO:0000471) is_a protein and gene product of some HLA-A gene

Figure 4. Antibody mediated MHC Identification and Validation.
Antibodies specific for one or more MHC molecules are used to isolate particular MHC molecules prior to epitope elution and that specificity is used to assert the identity of the MHC molecule presenting that epitope. The specificity for each antibody is known and is incorporated into the evidence through logical axioms in ECO, OBI, and MRO. That logic can then be used to identify curation errors in the IEDB, for example, when the wrong antibody:MHC pair was entered. In those cases, the original manuscript is reviewed to determine which of the two fields was in error and to correct the error. The same logic can be applied to prevent future errors.

OUTCOMES

The 42 IEDB MHC evidence terms mapped to 41 new ECO terms, because the logical modeling identified similarities between 2 terms, which collapsed into a single ECO term. This collaboration resulted in the creation of 144 new ECO terms, which includes organizational nodes and the creation of both manual assertion and automatic assertion terms for each type of evidence. A new search interface that allows users to narrow data based upon evidence was added to the IEDB website. Additionally, links to ECO were incorporated into the IEDB's webpages and data exports. This work also required the creation of 32 new OBI assay terms, used in the modeling of the methodology used to assert MHC evidence. The logical annotations of the ECO terms were used to identify errors and enforce inter-field validation of MHC Restriction and MHC Evidence fields in the IEDB. Autocompletion of the MHC term field based upon the evidence is now possible and is being considered as a future curation site enhancement.
This project illustrates how direct collaboration between groups using ontologies and those creating the ontologies provides mutual benefits.

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